Restaurant Module — Design Brief

Spa & Hospitality System Integration

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1. Overview & Scope

This module serves as the restaurant management component embedded within a broader spa and hospitality platform. It operates as a self-contained unit accessible to two distinct user groups — **guests** and **employees** — each with tailored interfaces, permissions, and workflows.

The module must seamlessly adapt to both full-service resort environments and day-use spa visitors, accommodating diverse dining needs ranging from formal table reservations to room service deliveries and packed trail lunches.

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2. User Roles & Interface Architecture

2.1 Guest Interface

The guest-facing interface is designed for clarity and minimal friction. A guest can access the module via the resort's digital touchpoints — personal devices, in-room tablets, or lobby kiosks — and perform all dining-related actions independently.

Core guest capabilities include browsing the menu, reserving a seat, pre-ordering food, viewing their assigned seating location on an interactive map, and requesting assistance. The interface must be welcoming, visually clean, and operable without any prior knowledge of the system.

2.2 Employee Interface

The employee-facing interface is a management tool designed for speed and accuracy in a fast-paced hospitality environment. Staff members can create and modify reservations on behalf of guests, access complete profiles including dietary data and package entitlements, view the live seating map, and coordinate food delivery to the correct seat or room.

A core design requirement is that employees can act as a guest's representative throughout the entire dining journey — from booking through to service completion.

2.3 Kiosk & Hardware Integration

The module must support physical kiosk terminals placed at restaurant entry points. Guests can tap their room keycard, wristband, or NFC-enabled device against a kiosk reader to instantly retrieve their booking, display their assigned seat on screen, and confirm arrival. This touchless, card-based check-in reduces queues and eliminates the need for a host to manually locate bookings.

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3. Reservation System

3.1 Seating Reservation — Guest-Initiated

Guests can browse available time slots, select their preferred seating area (indoor, outdoor, poolside, etc.), and confirm a reservation directly from the app or kiosk. The system provides real-time availability feedback and issues a digital confirmation with seat reference.

3.2 Seating Reservation — Employee-Initiated

Staff members can create reservations on behalf of walk-in guests or guests who have called ahead. The employee interface provides access to the full seating grid and allows manual assignment of seats, including accommodations for special needs such as wheelchair access or family seating clusters.

3.3 Interactive Seating Map

A live floor plan of the restaurant and surrounding dining areas is displayed as an interactive map. Available, occupied, and reserved seats are visually distinguished. Guests can select their preferred seat or view their already-assigned location. Employees use the same map to monitor the dining room at a glance and direct service efficiently.

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4. Food Ordering

4.1 Pre-Order from Menu

Guests and employees can browse the current menu and place orders ahead of the scheduled dining time. Menu items are presented with descriptions, allergen flags, and dietary labels. Pre-ordering allows the kitchen to prepare in advance, reducing wait times and enabling personalisation based on guest preferences.

4.2 Dietary Restrictions — Automatic Pull from Booking

When a guest's booking profile includes dietary restrictions or food allergies — entered at the point of accommodation booking — these are automatically surfaced throughout the ordering flow. Items incompatible with the guest's profile are clearly flagged or filtered by default. This data is also visible to employees when managing that guest's order.

4.3 Package Recognition — All-Inclusive vs. Pay-Per-Item

The system reads the guest's accommodation package to determine their meal entitlements. Guests on an all-inclusive plan proceed through the ordering flow without payment prompts; charges are handled behind the scenes. Guests on a standard plan see itemised pricing throughout and any outstanding food charges are automatically appended to their hotel bill upon checkout. The billing logic is transparent and requires no manual intervention from staff.

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5. Dining Contexts & Special Use Cases

5.1 External Diners (Non-Resident Guests)

The module supports walk-in or pre-booked guests who are not staying at the resort or hotel. These guests are identified in the system as external diners and are subject to standard payment flows at the point of ordering or upon departure. Their profiles do not carry accommodation-linked data, but they can still benefit from the full reservation and ordering experience.

5.2 Room Service

Hotel guests can place food orders for delivery directly to their room. The room service workflow guides the guest through the menu, confirms the delivery address (auto-populated from their room number), allows them to specify a delivery window, and tracks the order status in real time. Employees view room service orders in a dedicated queue separate from in-restaurant orders.

5.3 Families with Young Children

The reservation and ordering flows include a family-friendly mode that surfaces age-appropriate menu items, high chair availability, and allergy-aware children's options. When a family flag is present in the booking, the system can automatically reserve appropriately positioned seating (near restrooms, away from high-traffic zones) and prompt staff to prepare relevant amenities in advance.

5.4 Packed & Dry Food Takeaway

Guests planning excursions such as hiking, cycling, or day trips can pre-order packed meals or dry food parcels for collection at a scheduled time. The takeaway option presents a curated selection of portable, non-perishable, or ambient menu items. Orders can be placed the evening before for morning collection, ensuring minimal wait time at departure.

5.5 Assisted Dining & Accessibility Requests

Guests requiring assistance — whether due to mobility limitations, visual impairments, or other needs — can specify the type of support required at the time of booking or through the app. Supported requests include guided escorting to the seating area, assistance with ordering, adapted service formats such as tactile menus or verbal order confirmation, and assistance with eating if required. These requests are routed to the appropriate staff member and appear prominently in the employee interface to ensure no request is overlooked.

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6. Navigation — Interactive Route to Seating

When a guest has a confirmed reservation, the app provides an interactive map showing the route from their current location (room, spa, lobby, or entrance) to their assigned dining area. The map includes indoor wayfinding markers, points of interest, and estimated walk time. This feature reduces confusion in larger resort properties and enhances the overall guest experience without requiring staff intervention.

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7. Multi-Language Support

The full guest interface — including all menus, booking flows, wayfinding, and notifications — is available in multiple languages. Language preference is detected from the guest's booking profile or device settings, with the option to manually switch at any point. Employee interfaces support a separate configurable language, appropriate to the property's staff composition. All dynamic content (menu items, allergen labels, confirmation messages) must be translation-ready through a localisation management system.

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8. Integration Requirements

8.1 Property Management System (PMS)

The module connects to the hotel's PMS to retrieve guest profiles, room assignments, package types, check-in/check-out status, and dietary data captured during booking. Changes in guest status — such as early checkout or room upgrade — should reflect automatically within the restaurant module.

8.2 Point-of-Sale & Payment Hardware

For external diners and pay-per-item guests, the module integrates with card readers and POS terminals. In-app payment, card-on-file, and contactless hardware payments are all supported. The system is designed with an open integration layer so that property-specific payment hardware can be connected without custom development.

8.3 Kiosk Hardware & NFC/RFID Devices

Kiosk terminals communicate with the module's API to retrieve guest bookings in real time when a card or wristband is presented. The integration supports NFC, RFID, and magnetic stripe cards. The kiosk display is configurable and can show a personalised welcome screen, the guest's seat number, and any special instructions relevant to their visit.

8.4 Kitchen Display System (KDS)

Food orders placed through the module are pushed directly to the kitchen display in structured, prioritised format. Orders include dietary flags, special instructions, expected service time, and whether the order is for in-restaurant dining, room service, or takeaway collection.

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9. Design Principles

- **Clarity over density** — both interfaces should present information progressively, surfacing only what the user needs at each step of their journey.
- **Accessibility first** — all UI components must meet WCAG 2.1 AA standards, with particular attention to colour contrast, touch target sizing, and screen reader compatibility.
- **Consistency across surfaces** — the visual language, terminology, and interaction patterns should be consistent whether the guest is on a personal device, an in-room tablet, or a lobby kiosk.
- **Graceful degradation** — if a backend service is unavailable (PMS, KDS, or payment), the module should remain partially functional and communicate clearly to the user rather than failing silently.
- **Staff efficiency** — the employee interface is a working tool, not a showcase. Density, keyboard navigation, and quick-access patterns take priority over visual richness.

This brief is intended as the foundational specification for UX and UI design work. Each section corresponds to a distinct design workstream and can be developed and reviewed independently before integration.